ELEVENLABS HACKATHON · SPECIAL TRACK • VOICE AI INTEGRATION

AI Roundtable

A Real-Time Multi-Agent Podcast with Live Human Interruption

Autonomous AI personas collaborate, debate, and clash in broadcast-grade audio—powered by low-latency ElevenLabs speech synthesis and real-time human conversational steering.

VOICE AI ENGINE

ElevenLabs Turbo v2.5

Sub-second streaming TTS with custom prosody & distinct timbre per agent.

ORCHESTRATION

Dialectical Multi-Agent

Elena (Host), Marcus (Optimist), and Sophia (Skeptic) in fast 35-word banter.

CORE INNOVATION

Instant Interruption

Halts speech at 0ms, marks transcript, injects context, and dynamically pivots.

PROJECT TEAM

Shubham Raj • Rishabh Choudhary • Priyam Garg

THE MOTIVATION

Why Traditional AI Audio Falls Flat

⚠️ Status Quo Limitations

Robotic Monologues: LLMs generate essay-length monologues that bore audio listeners.

Zero Tension: Single-agent outputs lack authentic debate, skepticism, or counterarguments.

Passive Uninterruptible Audio: If a voice AI starts speaking, listeners are trapped with no way to jump in.

Uniform Timbre: One voice reading multiple viewpoints lacks broadcast credibility.

✨ The AI Roundtable Approach

Punchy Conversational Cadence: Strict 2–4 sentence (35–50 word) turn limits mimicking real podcast flow.

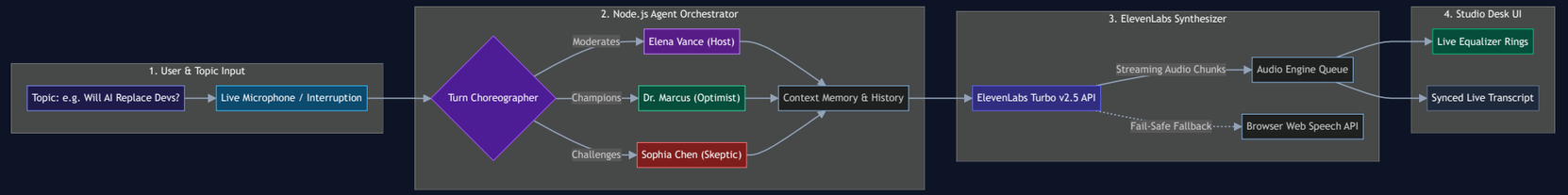
Autonomous Persona Friction: Optimist and Skeptic clash; Moderator balances and probes weak assumptions.

Real-Time Human Interruption: Live microphone STT halts speech instantly and prompts panelists to address you.

Distinct ElevenLabs Personas: Rachel, Adam, and Nicole give each personality authentic vocal depth.

SYSTEM ARCHITECTURE

End-to-End Multi-Agent Voice Pipeline



Orchestration Layer

Maintains conversational state, dynamic speaker selection, and persona prompts.

Voice AI Layer

Streams ElevenLabs MP3 binary chunks directly to browser AudioContext.

Interruption Layer

Web Speech STT with instant audio cancelation and context injection.

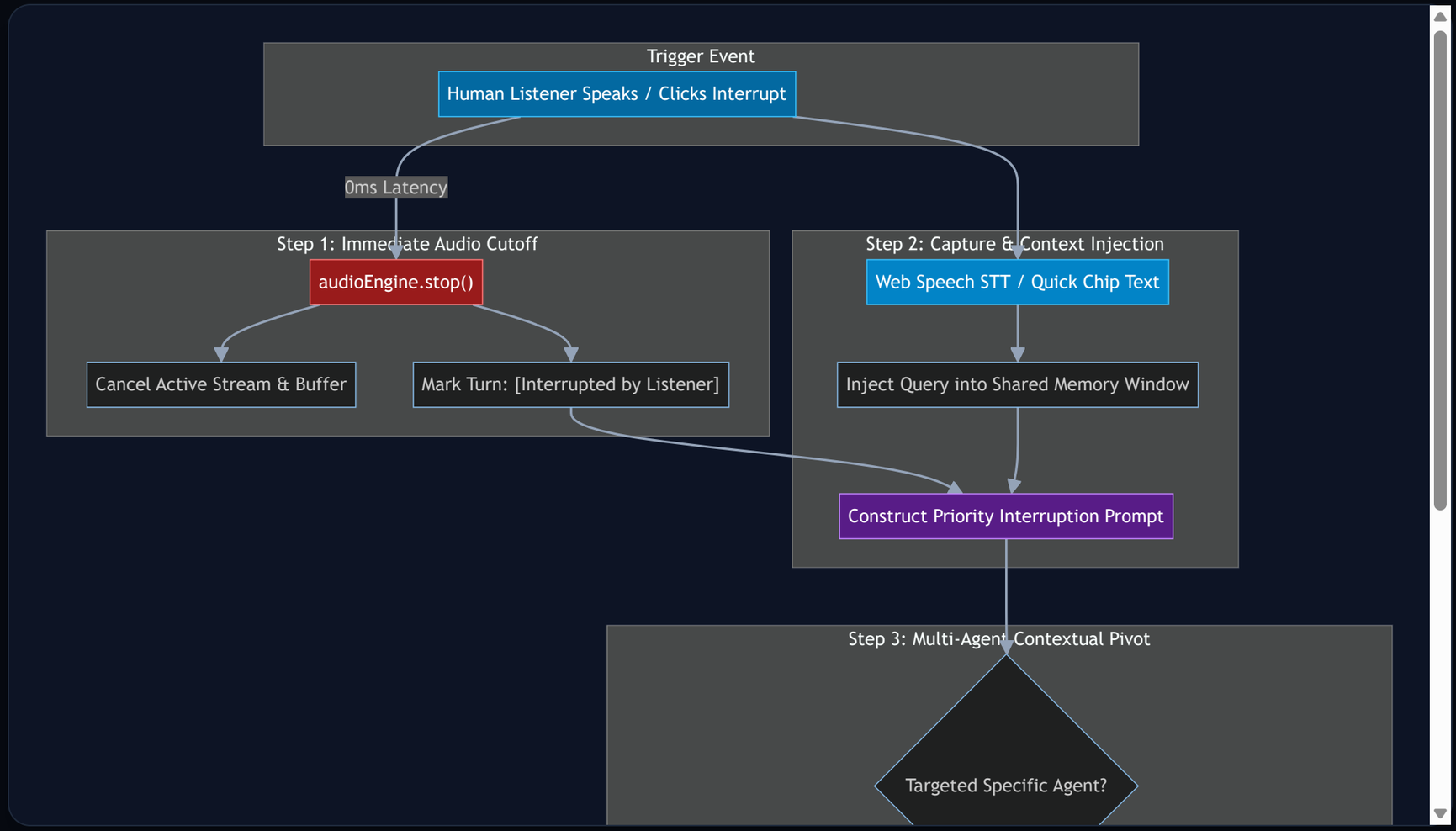
Fail-Safe Resilience

Zero stage crash guarantee: fallback speech synthesis if API quota is reached.

TECHNICAL SHOWCASE

Real-Time Human Interruption Lifecycle

Halting streaming speech instantaneously and dynamically pivoting the multi-agent context without repetition.



1. Stop Audio
Instant abort of active stream.

2. Mark State
[Interrupted] badge added.

3. Capture STT
Live Web Speech mic input.

4. Inject Context
Priority memory prompt.

5. Pivot & Resume
Agent addresses user by name.

PROOF OF CONCEPT

Live Demo: Broadcast-Quality Dialogue

EV

Elena Vance HOST · ELEVENLABS RACHEL

"Welcome to AI Roundtable. Today: 'Will AI replace software developers by 2028?'. Marcus, you claim this multiplies developer leverage. Start us off—what is your core thesis?"

MS

Dr. Marcus Sterling OPTIMIST · ELEVENLABS ADAM

"Thanks, Elena. Throughout history, every abstraction layer created more value. Compilers didn't eliminate coders; they allowed software to eat the world. Every engineer now commands a 10× leverage team!"

SC

Sophia Chen SKEPTIC · ELEVENLABS NICOLE

"Compilers were deterministic, Marcus. LLMs hallucinate false code. And if companies automate away junior roles, who trains the senior software architects of 2030?"

YOU

Audience Interruption (Shubham) LIVE MIC EVENT

[INTERRUPTS MARCUS] "Wait Marcus, what about the college grads who can't even get an entry-level interview today because companies expect 5 years of senior AI experience?"

MS

Dr. Marcus Sterling PIVOT REACTION

"Crucial point from our listener. While traditional junior tasks are shifting, software demand is effectively infinite. Apprenticeship will evolve into verification, system architecture, and domain modeling."

CONCLUSION & TEAM

Ready for Production & Judging

Shubham Raj

VOICE AI & FULL-STACK

Integrated ElevenLabs Turbo v2.5 streaming endpoints, built the Express backend orchestrator, and managed fail-safe audio buffers.

Rishabh Choudhary

AGENT PROMPTS & INTERRUPTION

Engineered dialectical role personas (Elena, Marcus, Sophia), turn arbitration heuristics, and dynamic listener acknowledgement prompts.

Priyam Garg

STUDIO UI & AUDIO REACTIVE ENGINE

Developed the circular studio desk, real-time equalizer waveforms, synchronized live transcript, and Web Speech STT integration.

Live Project Resources
Everything is open-source and running live on localhost & GitHub.

GitHub: shubhmroot/ai-roundtable ↗

Live Studio (Port 5173)